



# Lung Cancer Screening (PDQ®)–Health Professional Version

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## Overview

Note: The Overview section summarizes the published evidence on this topic. The rest of the summary describes the evidence in more detail.

Other PDQ summaries containing information related to lung cancer screening include the following:

- [Lung Cancer Prevention](#)
- [Small Cell Lung Cancer Treatment](#)
- [Non-Small Cell Lung Cancer Treatment](#)
- [Levels of Evidence for Cancer Screening and Prevention Studies](#)

## Evidence of Benefit Associated With Screening

### Screening by low-dose computed tomography (LDCT): Benefit

Two randomized trials have reported statistically significant reductions in lung cancer mortality associated with low-dose computed tomography (LDCT) screening. One trial reported that screening higher-risk individuals (30+ pack-years and either current smokers or quit within the past 15 years) aged 55 to 74 years three times, once annually, with LDCT reduced lung cancer mortality by 20% (95% confidence interval [CI], 6.8%–26.7%;  $P = .004$ ) and all-cause mortality by 6.7% (95% CI, 1.2%–13.6%;  $P = .02$ ) over screening with chest radiographs.[1] An updated analysis showed that the estimated reduction in lung cancer mortality was 16% (95% CI, 5%–25%).[2] The other trial reported that among high-risk current and former smokers, men who were randomly assigned to four rounds of LDCT screening had a 24% reduction (95% CI, 6%–39%) in lung cancer mortality, compared with men who were randomly assigned to no screening.[3]

**Magnitude of Effect:** About 20% to 24% relative reduction in lung cancer–specific mortality.

**Study Design:** Evidence obtained from randomized controlled trials (RCTs).

**Internal Validity:** Good.

**Consistency:** Good.

**External Validity:** Fair.

## Screening by LDCT: Harms

### False-positive exams

False-positive rates with LDCT screening have been high, although the magnitude of the rates varies with the definition of a positive screen.<sup>[1,4]</sup> False-positive exams may result in unnecessary invasive diagnostic procedures.

**Magnitude of Effect:** Two large randomized trials, the National Lung Screening Trial (NLST) and the Netherlands–Leuven Longkanker Screenings Onderzoek Trial (NELSON), found that the average false-positive rate per screening round was 23.3% and 10.4%, respectively.<sup>[1,3,4]</sup> Using a more recent definition of a positive LDCT screening on the basis of the Lung-RADS criteria yields a false-positive rate that may be somewhat lower than that seen in the NLST.<sup>[4]</sup> A total of 0.06% of all false-positive screening results in the NLST led to a major complication after an invasive procedure was performed as a diagnostic follow-up to the screening. Over three screening rounds, 1.8% of NLST participants who did not have lung cancer had an invasive procedure after a positive screening result.

**Study Design:** Evidence obtained from an RCT.

**Internal Validity:** Good.

**Consistency:** Good.

**External Validity:** Fair.

### Overdiagnosis from LDCT

Based on fair evidence, some lung cancers detected by LDCT screening appear to represent overdiagnosed cancer. However, estimates of overdiagnosis rates, derived typically by using data from randomized trials of LDCT screening, vary greatly. Therefore, the magnitude of overdiagnosis with LDCT screening is not clear. Overdiagnosed cancers result in unnecessary diagnostic procedures and also lead to unnecessary treatment. The harms of diagnostic procedures and treatment occur at the highest rate among long-term and/or heavy smokers because of smoking-associated comorbidities that increase risk propagation.

**Magnitude of Effect:** Uncertain.

**Study Design:** RCTs.

**Internal Validity:** Good.

**Consistency:** Evidence is consistent for the overall existence of overdiagnosis but is poor for determining the exact magnitude of effect.

**External Validity:** Fair.

## **Evidence of No Benefit Associated With Screening**

### **Screening by chest x-ray and/or sputum cytology: Benefits**

Based on solid evidence, screening with chest x-ray and/or sputum cytology does not reduce mortality from lung cancer in the general population or in ever-smokers.

**Magnitude of Effect:** Not applicable.

**Study Design:** RCTs.

**Internal Validity:** Good.

**Consistency:** Good.

**External Validity:** Good.

### **Screening by chest x-ray and/or sputum cytology: Harms**

#### **False-positive exams**

Based on solid evidence, false-positive rates with chest x-rays are in the range of 5% to 10% per exam. False-positive exams may result in unnecessary invasive diagnostic procedures.

**Study Design:** RCTs.

**Internal Validity:** Good.

**Consistency:** Good.

**External Validity:** Good.

#### **Overdiagnosis from chest x-ray and/or sputum cytology**

Based on fair evidence, some of the lung cancers detected by screening chest x-ray and/or sputum cytology appear to represent overdiagnosed cancer; however, the magnitude of overdiagnosis is not clear. These cancers result in unnecessary diagnostic procedures and also lead to unnecessary treatment. The harms of diagnostic procedures and treatment occur at the highest rate among long-term and/or heavy smokers because of smoking-associated comorbidities that increase risk propagation.

**Magnitude of Effect:** Uncertain.

**Study Design:** RCTs.

**Internal Validity:** Good.

**Consistency:** Evidence is consistent for the overall existence of overdiagnosis but is poor for determining the exact magnitude of effect.

**External Validity:** Good.

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## Incidence and Mortality

Lung cancer is the second most common form of noncutaneous cancer in the United States and is the leading cause of cancer death in both men and women. In 2025 alone, an estimated 110,680 men and 115,970 women will be diagnosed with lung cancer, and 64,190 men and 60,540 women will die of this disease. After rising rapidly over several decades in both sexes, the lung cancer death rate declined by 61% for men starting in 1990 and by 38% for women starting in 2002. From 2013 to 2022, death rates decreased by 4.8% per year in men and 3.7% per year in women.[1]

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## Risk Factors

The most important risk factor for lung cancer (as for many other cancers) is tobacco use.[1,2] Cigarette smoking has been definitively established by epidemiological and preclinical animal experimental data as the primary cause of lung cancer. This causative link has been widely recognized since the 1960s, when national reports in Great Britain and the United States brought the cancer risk of smoking prominently to public attention.[2] The percentages of lung cancers estimated to be caused by tobacco smoking in men and women are 90% and 78%, respectively.

For a complete description of factors associated with an increased or decreased risk of lung cancer, see [Lung Cancer Prevention](#).

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## Evidence of Benefit Associated With Screening

### Screening by Low-Dose Computed Tomography

There have been intensive efforts to improve lung cancer screening with newer technologies, including low-dose computed tomography (LDCT).[1,2] LDCT was shown to be more sensitive than chest radiography. In the Early Lung Cancer Action Project (ELCAP),[2] LDCT detected almost six times as many stage I lung cancers as chest radiography, and most of these tumors were no larger than 1 cm in diameter.

A systematic analysis [3] summarized 13 observational studies of LDCT, which included 60 to 5,201 participants and were conducted between 1993 and 2004. Some Japanese studies included nonsmokers, but the other studies were limited to current and former smokers. Variability in detection of nodules—between 3% and 51%—may be attributed to several factors:

- The definition of nodules (some studies required a size threshold).
- The computed tomography (CT) technology (thin slice detects more and smaller nodules).
- Geographic variation in endemic granulomatous disease.

Overall, lung cancer was diagnosed in 1.1% to 4.7% of screened participants. Most of these diagnoses were early-stage disease.[3]

The National Lung Screening Trial (NLST) provided the first solid evidence that screening with LDCT can reduce lung cancer mortality risk in ever-smokers who have smoked 30 pack-years or longer and in former smokers who have quit within the past 15 years. The NLST included 33 centers across the United States. Eligible participants were aged of 55 years to 74 years at randomization, had a history of at least 30 pack-years of cigarette smoking, and, if former smokers, had quit within the past 15 years. A total of 53,454 individuals were enrolled; 26,722 participants were randomly assigned to receive screening with LDCT, and 26,732 participants were randomly assigned to receive screening with chest x-ray. Any noncalcified nodule found with LDCT that measured at least 4 mm in any diameter and any noncalcified nodule or mass identified on x-ray images were classified as positive. Radiologists, however, had the option of calling a final screen negative if a noncalcified nodule had been stable on the three screening exams. The LDCT group had a substantially higher rate of positive screening tests than did the radiography group (round 1, 27.3% vs. 9.2%; round 2, 27.9% vs. 6.2%; and round 3, 16.8% vs. 5.0%). Overall, 39.1% of participants in the LDCT group and 16.0% in the radiography group had at least one positive screening result. Of those who screened positive, the proportion with lung cancer (i.e., positive predictive value) was 3.6% in the LDCT group and 5.5% in the radiography group.[4]

In the LDCT group, 649 cancers were diagnosed after a positive screening test, 44 after a negative screening test, and 367 among participants who either missed the screening or received the diagnosis after the completion of the screening phase. In the radiography group, 279 cancers were diagnosed after a positive screening test, 137 after a negative screening test, and 525 among participants who either missed the screening or received the diagnosis after the completion of the screening phase. Three hundred fifty-six deaths from lung cancer occurred in the LDCT group, and 443 deaths from lung cancer occurred in the chest x-ray group; the relative reduction in the rate of death from lung cancer was 20% (95% confidence interval [CI], 6.8%–26.7%) with LDCT screening at a median duration of follow-up of 6.5 years. [4] An updated analysis showed that the estimated reduction in lung cancer mortality was 16% (95% CI, 5%–25%).[5] Overall, mortality was reduced by 6.7% (95% CI, 1.2%–13.6%). The number needed to screen with LDCT to prevent one death from lung cancer was 320.[4]

An extended follow-up analysis of the NLST reported mortality data after a median of 12.3 years of follow-up. The estimated number needed to screen with LDCT to prevent one lung cancer death was 303.[6]

Since the publication of the results of the NLST, more has been learned about who may benefit the most from screening for lung cancer using LDCT.[7-9] One group of investigators developed an individual risk model to assess who might benefit from screening. The model



used additional factors not used as inclusion criteria in the NLST, such as a history of chronic obstructive pulmonary disease, personal or family history of lung cancer, and a more detailed smoking history. More individuals would have been eligible to be screened using the trial's criteria as opposed to the inclusion criteria of the NLST without missing patients with cancer.[8] A second group performed a reanalysis of the NLST data, calculated each patient's risk of developing lung cancer, and estimated each patient's lung cancer mortality.[9] The investigators then divided the NLST participants into five groups on the basis of risk. The number needed to screen to avoid a lung cancer death in the low-risk group was 5,276; 161 screens were needed in the high-risk group to avoid a lung cancer death. Furthermore, the number of false-positive screens decreased from 1,648 in the lowest quintile of risk to 65 in the highest-risk group. The three highest quintiles of risk accounted for 88% of the mortality reduction from screening, whereas the lowest quintile accounted for only a 1% reduction in mortality. These studies illustrate possible improvements for determining the population of patients who may benefit the most from screening, potentially reducing the number of false positives and reducing the potential harm related to the adverse events associated with their evaluation. One other benefit of calculating individual risk is the ability to incorporate the findings into a shared decision-making process so that patients can decide whether to undergo screening.[9] However, a comparison of ten models used for predicting lung cancer or lung cancer mortality risk found that four of the models were well calibrated with reasonable discrimination, but none of these models were considered superior to the others for identifying lung cancer risk among individuals who had ever smoked. Additional work is needed to address modeling weaknesses.[10]

The NELSON trial (Nederlands–Leuvens Longkanker Screenings Onderzoek) conducted in Belgium and the Netherlands examined screening for lung cancer in smokers (13,195 men, 2,594 women, and 3 unknown) with CT, using a volume criterion for positivity.[11] Participants were recruited from population registries in the two countries based on responses to questionnaires about their smoking history and other data. Those who either smoked currently or had quit for fewer than 10 years and had smoked more than 15 cigarettes a day for over 25 years or more than 10 cigarettes a day for over 30 years were eligible. Those with serious comorbidities or previous cancers were excluded. All participants were randomly assigned equally to either usual care or an initial screen and three subsequent screens at intervals of 1, 2, and 2.5 years. The screening test was LDCT, which was retrospectively analyzed by supervised software to determine nodule segmentation and volume. After a minimum follow-up of 10 years, 90% of men assigned to screening complied with each opportunity on average, with a 2.1% rate of being referred for diagnosis. These men experienced a lung cancer incidence rate of 5.58 per 1,000 person-years and a lung cancer–specific mortality rate of 2.5 per 1,000 person-years, compared with 4.91 cases and 3.3 deaths per 1,000 person-years in men assigned to usual care. The mortality rate ratio for screening was 0.76 (95% CI, 0.61–0.94). [11]

Although the NELSON study used a usual-care arm instead of a chest x-ray arm, the results are consistent with the main NLST results discussed above, both in the impact on lung cancer mortality and in overdiagnosis. The mortality results were even more similar when the NELSON cohort was constrained to the NLST smoking eligibility subgroup. The two studies diverged in several ways, however. The NLST observed an all-cause mortality reduction consistent with the dominant effect of lung cancer on mortality among smokers. NELSON did not find such an effect. In addition, no healthy volunteer effect was observed in NELSON, while the NLST reported a substantial effect. However, these differences between the studies do not cast doubt on the main effect on lung cancer mortality but may invite further analyses to understand the inconsistencies better.

Other, smaller randomized clinical trials (RCTs) of LDCT that compare a nonscreening arm with LDCT are under way or are already completed in a number of countries.[12-18] These smaller trials are not powered to assess mortality as an endpoint, but there is an effort to combine the findings from these studies with the NELSON data, once the data are fully mature. These studies may also assess consistency with the NLST findings. In addition to the data gleaned from ongoing trials, data from the NLST, NELSON, and other completed trials are being analyzed to examine other important issues in lung cancer screening, including cost-effectiveness, quality of life, and whether screening would benefit individuals younger than those enrolled in the NLST and those with fewer than 30 pack-years of smoking exposure. Data from the U.S. Prostate, Lung, Colorectal and Ovarian (PLCO) Cancer Screening Trial suggest that, in the absence of screening, the risk of lung cancer death for current smokers who have a smoking history of 20 to 29 pack-years is no different from that of former smokers who have quit within 15 years and have a smoking history of more than 30 pack-years (hazard ratio, 1.07; CI, 0.75–1.5).[19] Although the risk for the 20-to-29-pack-years current-smokers group is no different from that of the former-smokers group (for whom LDCT screening is recommended by the U.S. Preventive Services Task Force), the efficacy of screening is unknown in the 20-to-29-pack-years current-smokers group.[19]

*[Note: A [Guide](#) has been developed to help patients and physicians assess the benefits and harms of LDCT screening for lung cancer.[20](#)]*

## Screening and Smoking Cessation

The target population for lung cancer screening has a high prevalence of current smokers compared with the general population. A lung cancer screening program could potentially impact the likelihood of smoking cessation, theoretically promoting cessation among those screened who have lung abnormalities detected on their screen. Conversely, screening could also be a deterrent to cessation among those with no evidence of lung abnormalities on their screen. The Danish Lung Cancer Screening Trial is a randomized trial that compared LDCT with no intervention among participants aged 50 to 70 years who had at least a 20 pack-year



smoking history.[21] The proportion of participants who had quit smoking was monitored every year for 5 years of follow-up and remained virtually identical in the two groups from baseline (CT group and control group each had 23% ex-smokers) until the 5-year follow-up (43% ex-smokers in both groups). The comparison of these two randomized groups indicates that the CT screening program had zero net effect on the likelihood of smoking cessation.

Another report used data from the NLST to address the question of whether the screening result influenced the likelihood of smoking cessation.[22] The NLST compared CT with chest x-ray, and data from both arms were pooled to examine the impact of abnormal findings on the likelihood of smoking cessation. Compared with those who did not have abnormal findings, current smokers who had a screening examination that was suspicious for lung cancer (but was not lung cancer) were significantly more likely to have stopped smoking 1 year later. The associations with quitting smoking among those who had a major lung abnormality that was not suspicious for lung cancer, or those who had a minor abnormality, were weaker and not uniformly statistically significant.

A third study from the U.K. Lung Cancer Screening pilot trial of an LDCT scan found that screening was associated with a statistically significant increase in short- and long-term cessation, and this effect was greatest among those whose initial screening test was positive, warranting additional clinical investigation.[23]

The results of these studies suggest that the net impact of a CT program on smoking cessation varied,[21] but there appears to be a higher likelihood of smoking cessation among current smokers who have findings suspicious for lung cancer.[22] This is an important research area that needs to be clarified.

A meta-analysis that includes 85 RCTs published between 2010 and 2017 [24] concluded that electronic/Web-based, in-person counseling, and pharmacotherapy treatment interventions significantly increased the odds of successful smoking cessation among populations eligible for lung cancer screening.

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## Evidence of No Benefit Associated With Screening

### Screening by Chest X-ray and/or Sputum Cytology

The question of lung cancer screening dates back to the 1950s, when rising lung cancer incidence and mortality rates indicated a need for intervention. In response to the emerging lung cancer problem, five studies of chest imaging, two of which were controlled, were undertaken during the 1950s and 1960s.[\[1-8\]](#) Two studies also included sputum cytology.[\[1-5\]](#) The results of these studies suggested no overall benefit of screening, although design limitations prevented the studies from providing definitive evidence.

In the early 1970s, the National Cancer Institute funded the Cooperative Early Lung Cancer Detection Program,[9] which was designed to assess the ability of screening with radiologic chest imaging and sputum cytology to reduce lung cancer mortality in male smokers. The program comprised three separate randomized controlled trials (RCTs), each enrolling about 10,000 male participants aged 45 years and older who smoked at least one pack of cigarettes a day in the previous year. One study was conducted at the Mayo Clinic,[10-12] one at Johns Hopkins University,[13-15] and one at Memorial Sloan-Kettering Cancer Center.[15-18] The Hopkins and Sloan-Kettering studies employed the same design: participants randomly assigned to the intervention arm received sputum cytology every 4 months and annual chest imaging, while participants randomly assigned to the control arm received annual chest imaging. Neither study observed a reduction in lung cancer mortality with screening.[15] The two studies were interpreted as showing no benefit of frequent sputum cytology when added to an annual regimen of chest x-ray.

The design of the Mayo Clinic study (known as the Mayo Lung Project, or MLP), was different. All potential participants were screened with chest imaging and sputum cytology, and those known or suspected to have lung cancer, as well as those in poor health, were excluded. Remaining participants were randomly assigned to either an intervention arm that received chest imaging and sputum cytology every 4 months for 6 years, or to a control arm that received a one-time recommendation at trial entry to receive the same tests on an annual basis. No reduction in lung cancer mortality was observed. The MLP was interpreted in the 1970s as showing no benefit of an intense screening regimen with chest x-ray and sputum cytology.

One RCT of lung cancer screening with chest imaging was conducted in Europe in the 1970s. This Czechoslovakian study began with a prevalence screen (chest imaging and sputum cytology) of 6,364 men aged 40 to 64 years who were current smokers with a lifetime consumption of at least 150,000 cigarettes.[19,20] All participants except the 18 diagnosed with lung cancer as a result of the prevalence screen were randomly assigned to either an intervention arm or a control arm. Participants in the intervention arm received semiannual screening for 3 years. Participants in the control arm received screening during the third year only. The investigators reported 19 lung cancer deaths in the intervention arm and 13 in the control arm. They concluded that frequent screening was not necessary.

By 1990, the medical community was still unsure about the relationship between screening with chest imaging (using traditional chest x-ray) and lung cancer mortality. Although previous studies showed no benefit, findings were not definitive because of a lack of statistical power. A multiphasic trial with ample statistical power, the Prostate, Lung, Colorectal and Ovarian (PLCO) Cancer Screening Trial,[21] began in 1992. PLCO enrolled 154,901 participants aged 55 to 74 years, including women (50%) and never-smokers (45%). One-half of the participants

were randomly assigned to screening, and the other half of them were advised to receive their usual medical care. PLCO had 90% power to detect a 20% reduction in lung cancer mortality.

The lung component of PLCO addressed the question of whether annual single-view (posterior-anterior) chest x-ray was capable of reducing lung cancer mortality as compared with usual medical care. When the study began, all participants randomly assigned to screening were invited to receive a baseline and three annual chest x-ray screens, although the protocol ultimately was changed to screen never-smokers only three times. At 13 years of follow-up, 1,213 lung cancer deaths were observed in the intervention group, compared with 1,230 lung cancer deaths in the usual-care group (mortality relative risk, 0.99; 95% confidence interval, 0.87–1.22). Subanalyses suggested no differential effect by sex or smoking status.<sup>[21]</sup>

Given the abundance and consistency of evidence, as well as the lack of benefit observed in the PLCO trial, it is appropriate to conclude that lung cancer screening with chest x-ray and/or sputum cytology, regardless of sex or smoking status, does not reduce lung cancer mortality.

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## Harms of Screening



## Screening by Low-Dose Computed Tomography

### False-positive exams

False-positive exams are particularly problematic in the context of lung cancer screening. The individuals most likely to be screened for lung cancer, (i.e., heavy smokers) have comorbidities, such as chronic obstructive pulmonary disease and heart disease, that make them poor candidates for certain diagnostic procedures.

False-positive test results must be considered when lung cancer screening with low-dose computed tomography (LDCT) is being evaluated. A false-positive test may lead to anxiety and invasive diagnostic procedures, such as percutaneous needle biopsy or thoracotomy. The percentage of false-positive findings varies substantially among studies and is primarily attributable to differences in how a positive scan is defined (the size criteria), the thickness of the slice used between cuts (smaller slice thicknesses lead to detection of more nodules), and whether the subject resides in a geographic location where granulomatous disease is highly prevalent.

In the National Lung Screening Trial (NLST), the false-positive rate was 24% at baseline, and 27% and 16% for the two subsequent screening rounds.[1] In a systematic review of 20 studies (including the NLST), the median false-positive rate was 20.5% (range, 1%–49%) on baseline screens and 9.5% (range, 1%–42%) on postbaseline screens.[2] False-positive rates are generally lower on postbaseline screens because a nodule's growth rate can be assessed when there is a previous screen available, and stable (nongrowing) nodules are often denoted as negative screens. The Lung-RADs criteria for assessing LDCT findings, which are in wide use in the United States, are stricter than the NLST criteria for defining a positive screen, and have the potential to lower the false-positive rate from that seen in the NLST.[3]

### Diagnostic evaluations and downstream complications

A systematic review of the benefits and harms of computed tomography (CT) screening for lung cancer summarized 21 studies with respect to various diagnostic outcomes, although not all studies reported on all outcomes.[2] The rate of diagnostic CT imaging after a reported nodule varied from 0% to 45% of all individuals who were screened. Positron emission tomography scanning was performed in 2.5% to 5.5% of individuals who were screened. The frequency of nonsurgical biopsies or procedures ranged from 0.7% to 4.4% of individuals who were screened, with the finding of a benign result on biopsy ranging from 6% to 79%. The rate of surgical resection for screen-detected nodules was between 0.9% and 5.6% of individuals who were screened; the proportion among these with a benign result ranged from 6% to 45%.

In the NLST, most major complications were related to invasive procedures and surgeries performed on patients diagnosed with lung cancer, with a major complication rate of 11.8%.

The rates of complications from the NLST may not be generalizable to a community setting; participants in the NLST were younger, better educated, and less likely to be current smokers (therefore, healthier) than the population of smokers and former smokers in the general U.S. population who would be eligible for screening. Of note, 82% of the participants were enrolled at large academic medical centers, and 76% of the participants were enrolled at National Cancer Institute–designated cancer centers. However, diagnostic follow-up did not necessarily occur at the NLST screening centers and could have been carried out in community settings. This may account for the low complication rate and surgical mortality rate (1%) found in the NLST. These findings led the multisociety position paper to strongly recommend that screening be carried out at centers with the same patient-management resources as those in the NLST. [4]

A retrospective cohort study of community practices indirectly estimated the complication rates and downstream medical costs of invasive diagnostic procedures performed for lung abnormalities identified through lung cancer screening.[5] The observed complications rates of 22.2% (in patients aged 55–64 years) and 23.8% (in patients aged 65–77 years) were more than twice that reported in the NLST (8.5%–9.8%). The mean costs of managing complications ranged from \$6,320 (minor complication) to \$56,845 (major complication). These data suggest that the NLST, which was conducted in the context of a controlled clinical trial, may have underestimated the potential for adverse events and high downstream costs in the community setting. Study limitations include a lack of information about patient eligibility for lung cancer screening, the fact that the diagnostic procedures were not generally performed as follow-up to screening, and the extent to which complications were affected by poorer patient health and lower quality of care. Despite limitations, these results reinforce the need for the discussion about risks, benefits, and shared decision making.[5]

## Overdiagnosis

A less familiar harm is overdiagnosis, which means the diagnosis of a condition that would not have become clinically significant had it not been detected by screening [6]—that is, had the patient not been diagnosed with the cancer, the patient would have died of other causes. In the case of screening with LDCT, overdiagnosis could lead to unnecessary diagnosis of lung cancer requiring some combination of therapy (e.g., lobectomy, chemotherapy, and radiation therapy). Autopsy studies suggest that a significant number of individuals die with lung cancer rather than die *of* lung cancer. In one study, about one-sixth of all lung cancers found at autopsy had not been clinically recognized before death.[7] This may be an underestimate; depending on the extent of the autopsy, many small lung cancers that are detectable by CT may go unrecorded in an autopsy record.[8] Studies in Japan provided additional evidence that screening with LDCT could lead to a substantial amount of overdiagnosis.[9]

One approach to assessing overdiagnosis involves examining the volume-doubling time of lung tumors detected on LDCT. In one study, the volume-doubling times of 61 lung cancers were

estimated by using an exponential model and successive CT images. Lesions were classified into the three following types: type G (ground glass opacity), type GS (focal glass opacity with a solid central component), and type S (solid nodule).

The mean volume-doubling times were 813 days, 457 days, and 149 days for types G, GS, and S, respectively. In this study, annual CT screening identified a large number of slowly growing adenocarcinomas that were not visible on chest x-ray, suggesting overdiagnosis.[10]

In a screening cohort with more than 5,000 participants, volume-doubling time was used as a surrogate for overdiagnosis. Patients with a calculated volume-doubling time of more than 400 days before surgical resection were considered to have a slow-growing or indolent cancer.[11] The investigators discovered that 25% of incident cancers (31 of 120) met the criteria of a slow-growing or indolent tumor.[11] This rate is consistent with previous chest radiograph screening studies and for other solid tumors.

Another approach to assessing overdiagnosis is to compare lung cancer incidence rates across arms in randomized trials of LDCT screening. Data from the NLST showed a gap of about 120 excess lung cancer cases in the LDCT group compared with the chest radiograph group after a medium follow-up of 6.5 years (i.e., 4.5 years after the last scheduled screen). This suggests that 18% of screen-detected lung cancers (N = 649) were overdiagnosed.[12] However, an extended follow-up analysis of the NLST based on a median of 11.3 years of follow-up for incident cancer found a much smaller, and nonstatistically significant, excess of only 20 cancers in the LDCT group, resulting in an estimate of the percentage of overdiagnosed LDCT screen-detected cancers of 3%. Note that the NLST control group was screened with chest x-rays, so technically the above overdiagnosis estimates were in comparison with what would have been diagnosed with chest x-ray screening, not with what would have been diagnosed with no screening.

Additional evidence of overdiagnosis with LDCT screening was observed in the randomized Danish Lung Cancer Screening Trial. At 10 years of follow-up (5 years after the last screening exam), almost twice as many lung cancers had been diagnosed in the screening group as in the control group: 5.1 vs. 2.7 cases per 1,000 person-years or 100 vs. 53 lung cancer cases in 4,104 total participants, respectively. Most of the lung cancers were early-stage adenocarcinomas, with no statistically significant difference in the number of stage III and IV cancers between the two groups.[13] Overdiagnosis was estimated at 67%.[14] In three other small trials of LDCT screening, one showed a borderline significant increase in lung cancer incidence in the LDCT versus the control arm ( $P = .04$ ), suggesting overdiagnosis, while there was no significant difference in lung cancer incidence across arms in the other two trials.[15-17] In the NELSON trial (Nederlands-Leuvens Longkanker Screenings Onderzoek), with 4.5 years follow-up after the last screen, the overdiagnosis rate was 19.7% (95% confidence interval [CI], -5% to 42%). [18]

The overdiagnosis estimates from the NLST are compared with what would have been diagnosed with chest x-ray screening; therefore, in order to interpret them, it is necessary to have an estimate of the level of overdiagnosis using chest x-ray screening, preferably, covering a time period and population similar to those in the NLST. Such an estimate comes from the U.S. Prostate, Lung, Colorectal and Ovarian (PLCO) Cancer Screening Trial of chest x-ray screening versus usual care, specifically in the subset of PLCO trial participants who met the NLST eligibility criteria. These data showed no evidence of overdiagnosis, with essentially equivalent numbers of diagnosed lung cancers in the chest x-ray and usual-care arms after 3 years of follow-up after the last scheduled screen (rate ratio, 1.00).[19]

A meta-analysis of overdiagnosis from six randomized controlled trials, including the NLST and the NELSON, showed an aggregate overdiagnosis rate of 0.30 (95% CI, 0.06–0.55). The overdiagnosis rate was defined as the difference across arms in incident lung cancers divided by the number of screen-detected cases in the LDCT arm. However, there was significant heterogeneity ( $P = .0001$ ) in the overdiagnosis rate across trials, with two small trials showing rates around 0.65 and the NLST showing a low rate of 0.04.[20]

## Radiation exposure

Another potential risk from screening with LDCT is radiation exposure. The average exposure is low; the mean effective dose for LDCT in the NLST was 1.4 (SD = 0.5) mSv. It is estimated that over a 3-year period of screening, NLST participants were exposed to an average of 8 mSv of radiation (which accounts for radiation from screens and additional imaging for screen-detected nodules). A study of LDCT screens that were performed on more than 12,000 patients from 2016 to 2017 at 72 U.S. institutions found a mean effective dose of 1.2 (SD = 1.1) mSv. Almost two-thirds (65%) of the institutions had a median effective dose higher than the American College of Radiology guideline of 1 mSv. Modeling from previous work on radiation exposure and the development of cancer suggests that there could be one death per 2,500 screens in those participating in a screening program such as the NLST, although the benefit of screening of about one death avoided per 960 screens substantially outweighs the risk. Younger individuals and those without a significant risk of lung cancer may be more likely to suffer a radiation-induced lung cancer from screening than to be spared a lung cancer death. [2]

## Screening by Chest X-ray and/or Sputum Cytology

### False-positive exams

In the PLCO Cancer Screening Trial, the false-positive rate with chest x-ray screening ranged from 6.8% to 8.7% per exam over the four screening rounds.[21] In the NLST chest x-ray arm, false-positive rates were generally similar (range, 4.7%–8.7% over three rounds).[1]

## Diagnostic evaluation and downstream complications

In the NLST chest x-ray arm, among subjects with positive screens at baseline, 86% received imaging as diagnostic follow-up, 5% received a bronchoscopy, and 5% underwent a surgical procedure. Diagnostic imaging rates were modestly lower after postbaseline positive screens, while bronchoscopy and surgery rates were similar. A total of 0.3% of false-positive screens were associated with a complication of an invasive diagnostic procedure.[1]

In the PLCO trial, 0.4% of participants with at least one false-positive screen who had a diagnostic evaluation had a complication associated with a diagnostic procedure.[19] The most common of the 69 complications were pneumothorax (29%), atelectasis (15%), and infection (10%).

## Overdiagnosis

In the Mayo Lung Project trial of screening with chest x-ray and sputum cytology, after 5 years of follow-up after the last scheduled screen, 206 cancers were diagnosed in the screening arm compared with 160 cancers in the control arm.[22] Based on 90 screen-detected cancers in the screened arm, the overdiagnosis rate would be computed as 51% (i.e.,  $[206-160]/90$ ). After 13 years of follow-up in the PLCO trial, 1,696 lung cancers had been diagnosed in the intervention arm as compared with 1,620 cancers diagnosed in the usual-care arm, suggesting that about 25% of the 307 chest x-ray screen-detected cancers in the trial were overdiagnosed.[19] However, the incidence of lung cancer was not statistically different between the intervention and usual-care arms in the PLCO trial (rate ratio, 1.05; 95% CI, 0.98–1.12), indicating that the null hypothesis of no overdiagnosis could not be rejected.

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## Informed Medical Decision Making

Informed medical decision making is increasingly recommended for individuals who are considering cancer screening. Many different types and formats of decision aids have been studied. For more information, see [Cancer Screening Overview](#).

## Latest Updates to This Summary (04/17/2025)

The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

### Incidence and Mortality

Updated [statistics](#) with estimated new cases and deaths for 2025 (cited American Cancer Society as reference 1). Also revised text to state that after rising rapidly over several decades in both sexes, the lung cancer death rate declined by 61% for men starting in 1990 and by 38% for women starting in 2002. From 2013 to 2022, death rates decreased by 4.8% per year in men and 3.7% per year in women.

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## About This PDQ Summary

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